

# To accomplish the set goal, let's talk business & the revenue model

**Goal:** Increase the percentage of users who successfully retrieve a photo they remember but cannot precisely describe when they start searching.

## About & Mission

**Scope:** US & Indian Markets  
Primarily

### Mission:

*"Home for all your photos and videos"*

Google Photos is undergoing a foundational transition from a static cloud storage utility to an active, AI-driven autobiographical memory assistant.

## Strategy Shift

### 2015 Launch

Promise of "free unlimited storage" at compressed high quality.

### Today

Premium, AI-enhanced media management, positioning as a sophisticated intelligence layer over personal media.

## Revenue & Monetization

**150M+**

### Google One Subscribers

Primary acquisition funnel for Google One (Alphabet's overarching cloud subscription service).

**Generated \$2.64B** on Google Play Store alone in 2025.

**Resources:** <https://drive.google.com/file/d/1OWIOaCqmHv4bltepZDtv8doYQl26iTo0/view?usp=sharing>

# Let's look at ecosystem ties, insights, tradeoffs and outcomes

## Ecosystem Ties

Google Photos shares a unified 15GB free storage quota with Google Drive and Gmail. It also acts as a vital proving ground for multimodal Gemini AI models & Google Lens.

## Business Metric Decomposition

**Successful retrieval of vaguely remembered photos =**

**# Searches × # Queries per search × # Contextual photos retrieved per Query**

Contextual photos retrieved per Query  $\geq 1$  (if photo exists) to generate value for the user.

## Insights & Tradeoffs

With Google's introduction of "Ask Photos" powered by the Gemini 1.5 Pro model addresses semantic understanding, it suffers from critical latency trade-offs. The architecture currently struggles to meet the 300-millisecond threshold required to prevent user search abandonment, occasionally taking up to 2.5 seconds to process complex queries.

**Cloud vs. On-Device Tradeoff:** Relying on massive cloud models like Gemini for features such as Ask Photos introduces high latency and server costs, whereas utilizing on-device processing preserves privacy and battery but limits the depth of semantic understanding.

# Discovery-Engine findings and learnings

## Dataset

924 enriched entries  
from 11,807 raw review

## Sources

Play Store, App Store,  
Google Community

## Core Gap: Cognitive Mismatch

The engine identified 7 distinct opportunity areas from the 30 pain points.

Users remember photos through emotions, visual atmospheres, and contextual objects. The system is optimized for what users don't remember (exact dates, precise keywords) rather than what they do remember (feelings, people, vague contexts).

## Regional Insights

While US market was dominated by data loss trauma. For Indian Market memories are anchored to festivals, family gatherings, and seasons.

## 7 cognitive memory groups surfaced by discovery engine

| Memory Group             | Pain Pts |
|--------------------------|----------|
| Data Loss Trauma         | 13 (43%) |
| Temporal Amnesia         | 7 (23%)  |
| Sensory / Visual Cue     | 4 (13%)  |
| Event & Context          | 3 (10%)  |
| Facial & Identity Recall | 1 (3%)   |
| Lexical / Vocabulary Gap | 1 (3%)   |
| Pet & Object Confusion   | 1 (3%)   |

Link:

<https://google-photos-discovery-engine.vercel.app/>

# User problems, frictions, their needs and JTBD

**Sukriti** (26, Medical Practitioner)

## Study & Medical Reports

### Problem & Trigger:

Retrieving study screenshots (medical diagrams); hard to distinguish in text-heavy, identical piles.

### Friction & Need:

Scrolling un-contextual folders causes task abandonment. Desires search for medical entities.

**Aniket** (35, Travel Enthusiast)

## Travel & Trip Documents

### Problem & Trigger:

Navigating photo clusters on the spot (tickets, IDs, trip memories) without chronological anchors.

### Friction & Need:

Device migration broke metadata, making scrolling useless. Wants search to bypass timelines.

**Aarav** (32, Family-Oriented)

## Family Events & Collages

### Problem & Trigger:

Fatigue from multi-year infinite scroll when gathering family event photos for collages.

### Friction & Need:

Object tags work for simple items (bike), but fail complex event queries. Desires semantic search for broad events.

**JTBD Statement:** *"I want to describe the context, emotion, and background elements to the app in natural language, so it can instantly surface the exact memory I have in my mind, allowing me to relive, share that moment or use memory as utility when needed the most."*

# Defining target segment, writing user persona

## Segment — The Memory-Driven Travel Enthusiast

A casual-to-moderate Google Photos user who accumulates travel content organically - scenic shots, boarding passes, hotel confirmations, ID screenshots - across trips throughout the year, and later struggles to retrieve, organize, and repurpose that content for personal storytelling and social sharing.



**Priya** (28 yo)  
Marketing Manager,  
Bangalore  
Android (Pixel 8) • ~5k  
photos

*"I remember that beach. I wish I could search: Beach with big rocks"*



### Bio & Personality

**Bio:** Travels 2-3 times a year, capturing scenic moments, food, and travel documents in high bursts; never organizes them.

**Spectrum:** Extroverted, Spontaneous, Sentiment-driven, Visually-driven, Low-organization.



### Frustrations

Visual recall (e.g., "beach with big rocks") fails as a search query.



### Goals

- Build a year-end Instagram travel collage.
- Pull up IDs or boarding passes instantly at airport check-in.



### Behaviors & Habits

Gives up on search after 1-2 failed attempts, defaulting to stressful manual scrolling.

# Traversing user journey & identifying friction points to define the problem

## STEP 01

**Opens Google Photos app**

Intent to find photo

*"Let me find that gorgeous beach photo from Udupi"*

### Low Friction

Clear intent and effortless initial entry.

## STEP 02

**Scrolls the timeline**

*"There are way too many photos here, I'll never find it like this"*

### Low Friction

Standard interaction expected by user.

## STEP 03

**Taps search & types query**

**Irrelevant results**

*"May be if I type beach with big rocks..."*

### Medium Friction

Irrelevant results.



## STEP 04

**Tries different query**

**Still can't find it**

*"Udupi beach with hexagonal rocks"*

### High Friction

Partial results, the photo is missing.



## STEP 05

**Fallback to scrolling**

**Scrolls hundreds**

*Scrolls past ~1000 photos*

### High Friction

Memory is buried under noise.



## STEP 06

**Abandons the collage**

*"Forget it, I'll just post something random."*

### Critical Friction

Intent killed by retrieval friction.

# Now let's define the problem, document root cause and surface solution rationale by exercising Ideation

## What is the true problem?

When user searches for a photo they vividly remember, the app returns irrelevant results or nothing at all, leading them to believe their photos have been lost in the heap or deleted. This results in degraded trust.

## What is the value generated by solving this problem?

**For Customers:** They would own an autobiographical memory assistant.

**For Business:** The top of the funnel for Google One subscribers will have more leads.

## Root Cause

There is a cognitive mismatch between how the system indexes photos (structured metadata: dates, GPS, object tags) and how users recall them (sensory cues: scenes, emotions, visual details).

**Ideation:** A few possible solutions could be

- a) Surfacing an 'Add to Collection' or similar options to the users to enrich metadata.
- b) Provide user with filter options like shopping apps to filter on object tags.
- c) Associating a story to the photo and extracting tags from the story as metadata.

## Solution Rationale (Choosing option C)

As human own a story behind a memory, Wonder if AI could associate a brief story to an uploaded photo and extract sensory cue tags to enrich metadata. These tags then can be used to surface relevant photos. And this do not increase cognitive load.

## **MVP and user testing** Metrics, A/B Testing, Analysis, (Wireframes, User Flows, Prototype)

## Success metrics

## Risks and limitations